

Biomolecules

1. Which of the following statements regarding enzymes is incorrect?

- (1) Enzymes are bio catalysts that speed up biochemical reaction
- (2) Enzymes get consumed during the reaction and are not reusable
- (3) Each Enzyme are specific to a particular substrate
- (4) Enzymes lowers the Activation Energy of the reaction.

2. Given below are two statements

Assertion (A) : Molecular weight of lipid do not exceed 800 Da but it come under acid insoluble fraction.

Reason (R) : When we grind tissue cell membrane and other membrane are broken into pieces and form vesicles which are not water soluble.

Choose the correct answer from the given options :

- (1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (2) (A) is true but (R) is false
- (3) (A) is false but (R) is true
- (4) Both (A) and (R) are true and (R) is the correction explanation of (A)

3. Match the column-I to column-II :-

Column I		Column II	
A	Glutamic acid	(a)	Basic amino acid
B	Lysine	(b)	Acidic amino acid
C	Valine	(c)	Aromatic amino acid
D	Tryptophan	(d)	Neutral amino acid

- (1) i - c, ii - a, iii - b, iv - d
- (2) i - c, ii - a, iii - d, iv - b
- (3) i - c, ii - b, iii - d, iv - a
- (4) i - a, ii - b, iii - c, iv - d

4. True statements for cellulose molecule-

- (1) β -1-4linkage, unbranched
- (2) β -1-4linkage, branched
- (3) α -1-4 linkage, branched.
- (4) β -1-6linkage, unbranched

5. Choose the correct statement

- (i) All strictly macromolecules in living systems are polymers.
- (ii) Nucleic acids are composed of nucleotides.
- (iii) Polysaccharides occur only as energy storage molecules.

- (1) i and ii are correct, iii is incorrect
- (2) i and iii are correct, ii is incorrect
- (3) ii and iii are correct, i is incorrect
- (4) All three are correct

6. Select the option which is not correct with respect to enzyme action:

- (1) Substrate binds with enzyme at its active site.
- (2) Addition of lot of succinate does not reverse the inhibition of succinic dehydrogenase by malonate
- (3) A non- competitive inhibitor binds the enzyme at a site distinct from that which binds the substrate
- (4) Malonate is a competitive inhibitor of succinic dehydrogenase

7. Given below are two statements :

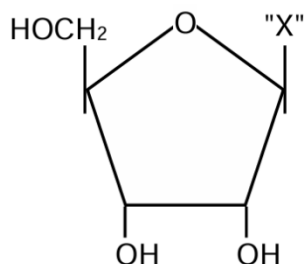
Statement I : Each enzyme shows it highest activity at optimum temperature and optimum pH.

Statement-II : Enzyme Isolated from organism who normal live under extremely high temperatures are stable and retain their catalytic power even at high temperature (upto 800C 900C).

Choose the correct answer from the option given below:

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- (4) Both Statement I and Statement II are correct

8. Given below is the diagrammatic representation of one of the categories of small molecular weight organic compounds in the living tissues. Identify the category shown and the one blank component "X" in it.



- | Category | Component |
|-----------------|-----------------|
| (1) Cholesterol | Guanine |
| (2) Amino acid | NH ₂ |
| (3) Nucleotide | Adenine |
| (4) Nucleoside | Uracil |
9. Cholesterol is an important compound in the body, because, It
- (1) regulates blood pressure
 - (2) gives rise to steroid hormones
 - (3) regulates filtration pressure in kidneys
 - (4) is involved in Ca⁺⁺ absorption from gut

10. Given below are two statements :

Statement A : Based on number of amino and carboxyl groups, amino acids are classified as acidic, basic and neutral amino acids.

Statement B : Glutamic acid is an example of basic amino acid.

Choose the correct answer from the options given below

- (1) Statement-A is correct but Statement-B is incorrect.
 - (2) Statement-A is incorrect but Statement-B is correct.
 - (3) Both Statements are correct.
 - (4) Both Statements are incorrect.
11. Which of the given statement is correct?
- (1) Lipids are generally water soluble
 - (2) Lipids could be with simple structures or complex structures
 - (3) Saturated fatty acids possess double bond between carbon atoms
 - (4) Lecithin is a glycolipid

12. Finds the correct matching, regarding secondary metabolites

- (1) Alkaloids – Morphine, Codien, Spices
- (2) Polymer – Cellulose, Gum, Rubber
- (3) Drugs – Vinblastin, Curcumin, Ricin
- (4) Pigments – Carotenoids, Anthocynin, Chlorophyll

13. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A) : Cellulose can not hold I₂ .

Reason (R) : Cellulose does not contain complex helices.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (2) (A) is correct but (R) is not correct
- (3) (A) is not correct but (R) is correct
- (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

14. How many of the given statements is/are incorrect?

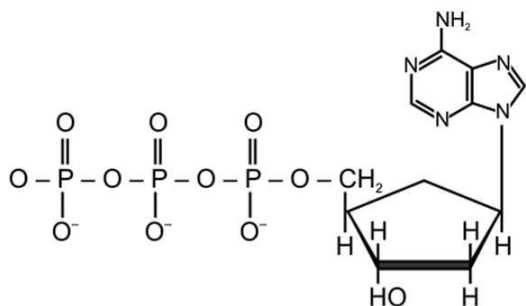
- a. A fatty acid has a carboxyl group attached to an R group
- b. R group could be methyl, ethyl or propyl only
- c. Palmitic acid has 16 carbons excluding carboxyl carbon
- d. Fatty acids could be saturated or unsaturated
- e. The fatty acids are found esterified with trihydroxy propane

- (1) 2
- (2) 3
- (3) 1
- (4) All are correct

15. Which of the following statement is correct about amino acid?

- (1) These are substituted methanes.
- (2) In these amino (–NH₂) and carboxyl (–COOH) group are of ionisable nature.
- (3) Based on the nature of R-group there are many amino acids
- (4) All of the above correct

16. The chemical structure shown in the figure is



- (1) a triphosphopeptide
- (2) a ribose sugar
- (3) deoxyadenosine triphosphate
- (4) the nitrogenous base, adenine

17. Find correct matching is/are :

- (a) Watson – Crick model = Secondary structure
- (b) Pocket of enzyme = 'Active site'
- (c) Trypsin = hormone

- (1) a and c only
- (2) a and b only
- (3) a only
- (4) a, b and c

18. Select the option which is not correct with respect to Co-factor.

- (1) Catalytic activity is lost when the co-factor is removed from the enzyme
- (2) Cofactor play a crucial role in the catalytic activity of the enzyme.
- (3) Three kinds of cofactors may be identified prosthetic groups, apoenzymes and metal ions.
- (4) Three kinds of cofactors may be identified: prosthetic groups, Co-enzymes and metal ions.

19. Consider the following statements:

Statement A - Coenzyme or metal ion that is tightly bound to enzyme protein is called prosthetic group,

Statement B - A complete catalytic active enzyme with its bound prosthetic group is called apoenzyme.

Choose the correct answer from the options given below

- (1) (A) is false but (B) is true
- (2) Both (A) and (B) are true
- (3) (A) is true but (B) is false
- (4) Both (A) and (B) are false

20. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason(R) .

Assertion: Enzymes are divided into different families .

Reason : All enzymes catalyse the same reactions.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.
- (2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
- (3) Assertion is true but Reason is false.
- (4) Both Assertion and Reason are false.

21. Match the following with respect to nucleic acid

Column I		Column II	
a.	Vinblastin	(i)	Alkaloid
b.	Ricin	(ii)	Polymer
c.	Codeine	(iii)	Drug
d.	Gum	(iv)	Poisonous

- (1) a-iii, b-iv, c- ii, d- i
- (2) a-i, b-ii, c- iv, d- iii
- (3) a-iii, b-iv, c- i, d- ii
- (4) a-iv, b-iii, c- i, d- iv

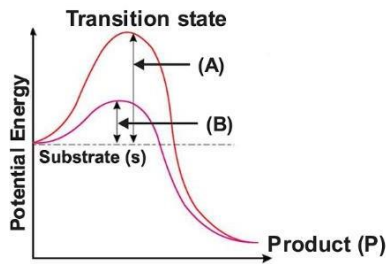
22. Transition state structure of the substrate formed during an enzymatic reaction is :

- (1) Transient and unstable
- (2) Permanent and stable
- (3) Transient but stable
- (4) Permanent but unstable

23. Select the option which is not correct with respect to Co-factor.

- (1) Catalytic activity is lost when the co-factor is removed from the enzyme
- (2) Cofactor play a crucial role in the catalytic activity of the enzyme.
- (3) Three kinds of cofactors may be identified prosthetic groups, apoenzymes and metal ions.
- (4) Three kinds of cofactors may be identified: prosthetic groups, Co-enzymes and metal ions.

24. In the given graphs which graphs represents, activation energy without enzyme and with enzyme. Choose the correct option :



- (1) A = Activation energy with enzyme
- (2) B = Activation energy with enzyme
- (3) B = Activation energy without enzyme
- (4) 1 and 3 both

25. Identify the correct set of statements out of followings :

- (a) C, H, O and N are present in proteins
- (b) Proteins are hetero polymers of amino acids
- (c) Glycine and serine are essential amino acids
- (d) Histidine is alkaline and glutamine is acidic in nature.

Choose the correct answer from the given options below

- (1) a, b, c and d
- (2) b, c and d
- (3) a, c and d
- (4) a, b and d

26. Which of the following structure is not common in all proteins?

- (1) Primary structure
- (2) Secondary structure
- (3) Tertiary structure
- (4) Quarternary structure

27. Given below are two statements :

Statement I : Each of metabolic reactions results in transformations of biomolecules

Statement II : Protein with cataytic power are named enzymes

Choose the correct answer

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- (4) Both, Statement I and Statement II are correct

28. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason(R) .

Assertion (A) : The lipids obtained from animals contain more quantity of saturated fatty acids.

Reason (R) : The lipids obtained from plants contain more quantity of unsaturated fatty acids.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) If (A) & (R) both are correct but (R) is not the correct explanation of (A).
- (2) (A) is correct but (R) is incorrect.
- (3) (A) is not correct but (R) is correct.
- (4) If (A) & (R) both are correct and (R) is the correct explanation of (A).

29. Chitinous exoskelton of arthrodopds is formed by the polymerisation of :

- (1) N – acetyl glucosamine
- (2) Lipiglycans
- (3) D – Glucosamine
- (4) Keraitin sulphate

30. Which of the following incorrect option :

- (1) Adenine, Guanine, Cytosine, Thymine → Nitrogen base
- (2) Nitrogen base + Sugar → Nucleoside
- (3) Nitrogen base + Sugar + Phosphate → Nucleotide
- (4) Nitrogen base + Phosphate → Nucleoside

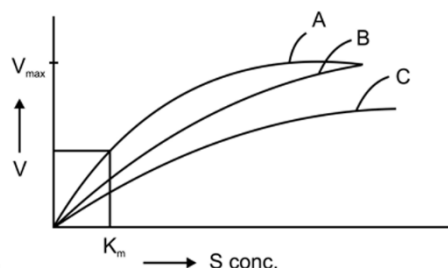
31. Which of the following statements is NOT CORRECT?

- (1) The main source of coenzymes are vitamins
- (2) Competitive inhibition of enzyme is irreversible
- (3) Enzymes have one or more active sites to interact with the substrate
- (4) Proenzymes have to be activated by proteolytic enzyme

32. Histone is a basic protein due to

- (1) Alanine & Glycine
- (2) Methionine & serine
- (3) Tryptophan & tyrosine
- (4) Lysine & Arginine

33. The figure given below shows three velocity substrate concentration curves for an enzyme reaction, What do the curves a, b, and c depict respectively?



- (1) a-normal enzyme reaction, b-competitive inhibition, c-non-competitive inhibition,
- (2) a-enzyme with an allosteric modulator added, b-normal enzyme activity, c-competitive inhibition
- (3) a-enzyme with an allosteric stimulator, b-competitive inhibitor added, c-normal enzyme reaction
- (4) a-normal enzyme reaction, b-non-competitive inhibitor added, c-allosteric inhibitor added

34. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason(R) .

Assertion (A) : Cholesterol is a type of derived lipids.

Reason (R) : Cholesterol is abundantly present in brain, nervous tissue, adrenal gland and skin.

Choose the most appropriate answer from the options given below :

- (1) If (A) & (R) both are correct but (R) is not the correct explanation of (A).
- (2) (A) is correct but (R) is incorrect.
- (3) (A) is not correct but (R) is correct.
- (4) If (A) & (R) both are correct and (R) is the correct explanation of (A).

35. Consider the following statement :

(1) Coenzyme or metal ion that is tightly bound to enzyme protein is called prosthetic group.

(2) A complete catalytic active enzyme with its bound prosthetic group is called apoenzyme.

Select the correct option.

- (1) Both (1) and (2) are true
- (2) (1) is true but (2) is false.
- (3) Both (1) and (2) are false.
- (4) (1) is false but (2) is true.

36. Regarding catalytic cycle of an enzyme action, select the correct sequential steps :

I. Substrate enzyme complex formation.

II. Free enzyme ready to bind with another substrate.

III. Release of products.

IV. Chemical bonds of the substrate broken.

V. Substrate binding to active site.

Choose the correct answer from the options given below :

- (1) V - I - IV - III - II
- (2) I - V - II - IV - III
- (3) II - I - III - IV - V
- (4) V - IV - III - II - I

37. Choose the correct statement(s) with respect to proteins.

- (1) In proteins, only right handed helices are observed
- (2) In the primary structure of a protein, the first amino acid is called C-terminal acid
- (3) Quaternary structures of proteins always involve two polypeptide subunits
- (4) All of the above

38. Read the following statements about the proteins

a. Proteins are the building blocks of life

b. Protein makes up the enzymes that power many chemical reactions

c. They do not help in tissue regeneration, defense activities

d. Proteins are micromolecules that comprises small chains of amino acids

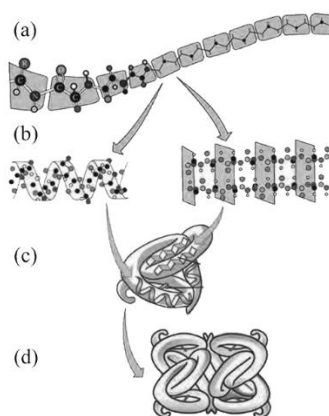
Choose the incorrect statements from the options given below :

- (1) b,d
- (2) d only
- (3) c,d
- (4) All are correct

39. Which of the following is not a similarity between starch and cellulose?

- (1) They are homopolysaccharides
- (2) They are found mainly in plants
- (3) They form helical structure and hold I_2
- (4) They have glycosidic bonds

40. Following diagram shows different levels of protein structure. Identify the labels a to d in the diagram?



- (1) a - Quaternary, b - Tertiary, c - Primary, d - Secondary
- (2) a - Secondary, b - Primary, c - Quaternary, d - Tertiary
- (3) a - Primary, b - Secondary, c - Tertiary, d - Quaternary
- (4) a - Secondary, b - Quaternary, c - Tertiary, d - Primary

41. Animal starch differs from plant starch in-

- (1) Having short chain branched structure
- (2) Being reserve food of animals mostly
- (3) Give no blue colour with iodine
- (4) All of the above

42. Match the column-I to column-II :-

Column I		Column II	
(A)	Heteropolymer of amino acids	(P)	Codes for transacetylase
(B)	Peptide bond	(Q)	Protein
(C)	Source of essential amino acids	(R)	Dietary proteins
(D)	Collagen	(S)	Abundant protein in whole of the biosphere
(E)	RuBisCO	(T)	Most abundant protein in animal world

- (1) A-Q, B-R, C-P, D-S, E-T
- (2) A-R, B-Q, C-T, D-P, E-S
- (3) A-Q, B-P, C-R, D-T, E-S
- (4) A-P, B-Q, C-R, D-S, E-T

43. Given below are two statements:

Statement-I : Acetone dihydroxy is a type of triose sugar.

Statement-II : Glycogen is animal starch which is a polymer of α -D glucose.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both Statement-I and II are correct.
- (2) Statement-I is correct but statement-II is incorrect.
- (3) Statement-I is incorrect and Statement-II is correct.
- (4) Both Statement-I and Statement-II are incorrect.

44. In the DNA molecule

- (1) the total amount of purine nucleotides and pyrimidine nucleotides is not always equal
- (2) there are two strands which run parallel in the 5' $\rightarrow$ 3' direction
- (3) the proportion of adenine in relation to thymine varies with the organism
- (4) there are two strands which run antiparallel-one in 5' $\rightarrow$ 3' direction and other in 3' $\rightarrow$ 5'

45. Which of the following statements is/are correct:

- (i) Tertiary structure is necessary for biological activity of protein
- (ii) Nucleic acids exhibit a wide variety of secondary structures
- (iii) Haemoglobin is an example of quaternary structure
- (iv) In primary structure of protein the first amino acid is called as N-terminal amino acid and the last amino acid is called C-terminal amino acid

- (1) ii only
- (2) i and iv only
- (3) i only
- (4) i, ii, iii and iv

46. Jute fibres deteriorate quickly because

- (1) cellulose content is high
- (2) lignin content is high
- (3) cellulose content is low
- (4) lignin content is low

47. Which one of the following correct statement :

- (a) Prosthetic group are organic compound
- (b) Proteineous portion of enzyme is called apoenzyme
- (c) Zn^{+2} is a co-factor for the proteolytic enzyme carboxypeptidase
- (d) NAD and FAD are Co-factor

- (1) Only a, b
- (2) Only a, d
- (3) Only c, d
- (4) a, b, c and d

48. Identify the correct set of statements out of followings :

- (a) Primary structure of protein is like a long straight chain
- (b) Secondary structure of protein looks like zig-zag
- (c) Tertiary structure of protein are insoluble in water
- (d) Maximum proteins present in protoplasm show secondary structure

Choose the correct answer from the given options below :

- (1) a, b and d
- (2) only a and b
- (3) b, c and d
- (4) a, b, c and d all

49. Given below are two statements :

Statement A : Low temperature preserves the enzyme in a temporarily inactive state whereas

high temperature destroys enzymatic activity because proteins are denatured by heat.

Statement B : When the inhibitor closely resembles the substrate in its molecular structure and inhibits the activity of the enzyme, it is known as competitive inhibitor.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both Statement A and Statement B are true.
- (2) Both Statement A and Statement B are false.
- (3) Statement A is true but Statement B is false.
- (4) Statement A is false but Statement B is true.

50. Match column I with column II and select the correct option from the codes given below.

Column I		Column II	
(i)	Collagen	(a)	Hormone
(ii)	GLUT - 4	(b)	Inter cellular ground substances
(iii)	Insulin	(c)	Sensory receptor
(iv)	Receptor	(d)	Enables glucose transport into cells

- (1) i-b, ii-d, iii-a, iv-c
- (2) i-b, ii-d, iii-c, iv-a
- (3) i-d, ii-b, iii-a, iv-c
- (4) i-c, ii-a, iii-d, iv-b

Answer Key

1. 2
2. 4
3. 2
4. 1
5. 1
6. 2
7. 4
8. 4
9. 2
10. 1
11. 2
12. 2
13. 4
14. 1
15. 4
16. 3
17. 2
18. 3
19. 4
20. 4
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22. 1
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26. 4
27. 4
28. 1
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30. 4
31. 2
32. 4
33. 1
34. 1
35. 2
36. 1
37. 1
38. 3
39. 3
40. 3
41. 4
42. 3
43. 1
44. 4
45. 4
46. 1
47. 4
48. 2
49. 1
50. 1